

Estimation of Mean using Empirical Maximum Likelihood

Basic Equations: Assume (for now) that we're working with a RV distribution of N integers. In the limit of large N , the data can be used to infer a set of probabilities p_i for each integer i . The basic equations are:

$$\begin{aligned}\sum_{i=1}^N p_i &= 1 \\ L &= \prod_{i=1}^N p_i \\ L_{MAX} &= \prod_{i=1}^N \frac{1}{N}\end{aligned}$$

If the mean of the data $\langle x \rangle = \mu_0$, we add the following constraint to the above system of equations:

$$\sum_{i=1}^N p_i (x_i - \mu_0) = 0$$

We employ the *Lagrange Multiplier* technique to extract the empirical probabilities $\tilde{p}_i$

$$\tilde{p}_i = \frac{1}{[1 + \lambda(x_i - \mu_0)]N}$$

where λ is the Lagrange Multiplier. The constraint equation becomes

$$\sum_{i=1}^N \frac{(x_i - \mu_0)}{[1 + \lambda(x_i - \mu_0)]} = 0$$

and we use this to determine λ .

Now, we can determine the mean μ_0 by maximizing L , or more typically

$$\begin{aligned}\frac{L}{L_{MAX}} &= \frac{\prod_{i=1}^N \tilde{p}_i}{\prod_{i=1}^N \frac{1}{N}} = \prod_{i=1}^N N \tilde{p}_i \\ \ln\left(\frac{L}{L_{MAX}}\right) &= -\sum_{i=1}^N \ln[1 + \lambda(x_i - \mu_0)]\end{aligned}$$

A useful result from Owen¹ is that the quantity

$$\begin{aligned}W &= -2 \ln\left(\frac{L}{L_{MAX}}\right) \\ &= 2 \sum_{i=1}^N \ln[1 + \lambda(x_i - \mu_0)]\end{aligned}$$

will converge to a χ^2 distribution with $N_{df} = (\text{number of constraints})$. As shown in the `MELE_example.ipynb` notebook, this is useful in computing confidence intervals from the empirical likelihood ratio (ELR).

¹Owen, A.B., "Empirical likelihood ratio confidence intervals for a single functional", *Biometrika* **75** (2): pp 237–249 (1988).

Generalization of Empirical Maximum Likelihood

In our evaluation of the mean, we expressed the constraint as

$$\langle x \rangle = \mu_0$$

or more suggestively as

$$\langle x - \mu_0 \rangle = 0$$

We can regard

$$g(x, \mu_0) = x - \mu_0$$

as an “estimating function” for the mean, in that setting it’s expectation value to zero

$$\langle g \rangle = 0$$

results in an expression for determining the mean.

We can write estimating functions for other quantities as well. This is particularly easy for the higher moments, which are defined as averages of some quantity over the entire data set. For example, an estimating function for the variance (in the case of a known mean) would be

$$g(x, \mu, \sigma^2) = (x - \mu)^2 - \sigma^2$$

The procedure for extracting a quantity from the data using the ELR method is just to replace $(x - \mu_0)$ on the previous page with the relevant estimating function g , in particular

The constraint equation:

$$\sum_{i=1}^N \frac{g}{[1 + \lambda g]} = 0$$

The ELR:

$$W = 2 \sum_{i=1}^N \ln [1 + \lambda g]$$

In today’s lab you will practice this procedure by extracting the variance and it’s 95% confidence interval from the Weibull distribution.